Did the 2025 US Government Shutdown Change Go-Around Rates? A Preregistered Study of Open Flight-Track Data at 76 Airports

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Abstract

During the lapse in US federal appropriations from 1 October to 12 November 2025 air traffic controllers worked without pay, and from 7 November the Federal Aviation Administration (FAA) ordered flight reductions at 40 airports, citing unpublished safety data. This preregistered study measured the go-around rate from open flight-track data for 77 days in 2025 and the same weekdays in 2024, at 50 US airports and 26 Canadian and European control airports (about 4.4 million approaches). At the US airports the rate was 4.77 per 1,000 approaches in the 2024 comparison window and 4.64 during the shutdown; at the control airports it was 3.44 and 3.66. The ratio of the US year-on-year change to the control change was 0.93 (95% interval 0.76 to 1.13). Under the preregistered rules this is inconclusive: no change was detected, but the interval is too wide to declare the absence of a meaningful one. The same comparison for the September before the shutdown, which should have shown nothing, gave 0.78 (0.66 to 0.93), and the same pattern ran through the first two weeks of October, so the two years were not on a common footing. Whole-window estimates that vary the reference period, the weather adjustment, the definition of a go-around and the airports included range from 0.81 to 1.19, and only one of their intervals excludes no change, from a weather model whose assumptions the data reject. No analysis shows a clear rise in go-arounds, and the study cannot exclude a moderate one. Two secondary measures gave apparent falls that did not survive review. The fall in collision-avoidance resolution advisories traces to a change in how the archive logged those messages in late September 2024 and to procedure changes at two airports. The fall in closely spaced landing pairs depends on how low the leading aircraft was tracked, which changed between the years. The study measures how often jets discontinued approaches. It does not measure losses of separation, runway incursions or controller workload.

1 Summary of findings

1. **Go-arounds did not detectably change.** During the shutdown window (1 October to 12 November 2025) the 50 US study airports recorded 3 997 go-arounds in 860 760 approaches, 4.64 per 1,000. On the same weekdays in 2024 the figures were 4 062 in 851 956, 4.77 per 1,000. The control airports went from 3.44 to 3.66 per 1,000. The preregistered estimate, the ratio of rate ratios, is 0.93 with a 95% interval of 0.76 to 1.13 (Table 1).

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2. **The preregistered verdict is "inconclusive".** The interval includes 1, so no change was detected. It extends below 0.80, so the rule for declaring "no meaningful change" is not met.

3. **A preregistered check failed, and it weakens the result.** The same comparison in September, before the shutdown, gave 0.78 (0.66 to 0.93). US go-around rates were higher in September 2024 than in September 2025 at 40 of the 50 US airports, and the gap continued through the first two weeks of October. Measured against the September relationship, the shutdown estimate is 1.19 (0.91 to 1.53). The upper limit of the primary interval, a rise of 13%, therefore holds only if the control airports are a fair comparison, which this check calls into question.

4. **The result is stable to how go-arounds are counted and which airports are included.** Estimates that drop low go-arounds, go-arounds linked to a collision-avoidance alert, airports with a runway present in only one year, or airports whose coverage changed lie between 0.90 and 1.08, and every interval includes 1 (Table 3).

5. **The study was less sensitive than planned.** The preregistration expected to detect a relative change of 21 to 23%. Day-to-day clustering was stronger than assumed. The realised figure is 33%, or 36 to 40% when the correlation between neighbouring days is allowed for.

6. **Single-airport changes are not shutdown effects.** Only Washington Dulles shows a year-on-year change that survives the preregistered correction for testing 76 airports. The largest other changes trace to a runway that was closed or not detected in one of the years, to a change of runway configuration, or to a fall in alert-related go-arounds at San Francisco.

7. **The two secondary measures are not interpretable.** Both gave statistically clear falls on the preregistered analysis. Review showed that each is driven by something other than the shutdown (Sections 4.3 and 4.4).

2 Background

A lapse in appropriations began on 1 October 2025 and ended on 12 November 2025. The FAA's emergency order of 7 November states that "air traffic controllers have been working without pay since October 3, 2025" [1]. The order required airlines to cut scheduled domestic flights at 40 "High Impact Airports" by 4% from 7 November, with further steps planned up to 10%. It also allowed air traffic control at short-staffed facilities to withhold services to aircraft flying under visual flight rules (VFR), including traffic-pattern operations and practice approaches [1]. A replacement order effective 13 November set the reduction at 6% and prohibited most general aviation at 12 of the airports [2]. The orders were cancelled with effect from 6:00 a.m. Eastern time on 17 November [3].

The first order justified the reductions by reference to safety data: the FAA had "assessed voluntary safety reports submitted for the month of October" and observed "evidence of increased stress on the NAS in aviation safety data", the NAS being the national airspace system [1]. The cancellation notice says that these data "indicated a correlation between ATO staffing and stress on the NAS" [3]. None of the three notices gives a number. A Congressional Research Service review published in October 2026 reports delay and cancellation counts for the shutdown but no safety-event data and cites no empirical safety study [4].

Published measurements of the shutdown concern delay and disruption. The Federal Reserve Bank of Richmond compared delays at 82 airports inside and outside the shutdown

window and found statistically significant increases at six: Boston, Washington National, Newark, New York JFK, LaGuardia and San Francisco [5]. A student report examined cancellations at Orlando [6]. Searches on 11 and 12 November 2026 found no study that measured a safety indicator for this period from flight data. The searches did not cover every index (Section 5).

Go-arounds can be detected in crowd-sourced surveillance data. Published rates from such data are a few per 1,000 approaches: about 2.5 per 1,000 at Mumbai [7], "one to three per 1000 commercial flight approaches" as a general figure quoted in a study of San Francisco [8], and 1.5 to 6 per 1,000 across 20 large European airports [9].

A go-around is a normal manoeuvre and is often the conservative choice. Its rate is used here as an indicator of how often an approach was discontinued, for whatever reason: spacing, an occupied runway, an unstable approach, weather, a collision-avoidance alert or an instruction from the tower. A lower rate is not in itself safer, since continuing an approach that should have been discontinued also lowers it. A change in either direction would be a signal to investigate and not a safety verdict.

What to expect. Short staffing is managed mainly upstream of the runway. Facilities reduce the rate at which they accept arrivals and impose miles-in-trail restrictions, ground delay programmes and ground stops, all of which lower pressure on final approach. Towers may combine positions, which could raise go-arounds for an occupied runway. Withholding services to VFR traffic reduces the number of light aircraft interacting with jets near the large airports. Many go-arounds are initiated by crews for weather or an unstable approach and should not respond to staffing at all. On that reasoning the expectation was a flat or slightly lower go-around rate, and the preregistration stated that a flat result was the most likely outcome and would be reported as such.

3 Method

The hypotheses, airports, windows, measures, exclusions, analysis and decision rules were fixed in a preregistration committed to a public repository on 11 November 2026, before any study-window data had been downloaded (`PREREGISTRATION.md`, commit `bfb0eae`). The time-stamp is that of the public commit; the document was not lodged with a registry. Section 3.6 lists every departure from it.

3.1 Data

Flight tracks came from the adsb.lol archive, which publishes one file set per day of all aircraft traces received by its volunteer network [10]. Seventy-seven days were processed for 2025 (1 September to 16 November) and 77 for 2024 (2 September to 17 November). Each 2024 day is the 2025 date minus 364 days, so the two years are aligned by day of the week. Days are UTC days. Every download was checked against the archive's published size; one oversized trace file was skipped and one file could not be read. Runway positions came from OurAirports [11]. Hourly weather reports (METAR) for every study airport came from the Iowa Environmental Mesonet [12].

Only aircraft whose type code is in fixed lists of airline jets and business jets were kept, and only positions within 15 NM of a study airport and below 7 000 ft above it.

3.2 Airports

Eighty-four airports were preregistered: the 40 airports of the FAA order, 15 other US airports, 4 Canadian and 25 European. A coverage rule fixed in advance removed any airport

that, in either year, had fewer than 60 of 77 days with at least 30 detected approaches. It removed eight: Houston Intercontinental, Columbus, Jacksonville, Kansas City, Pittsburgh, Vancouver, Oslo and Paris Orly. The analysis therefore covers 39 order airports, 11 other US airports, 3 Canadian and 23 European airports.

3.3 Measures

Go-around rate (primary). A landing is the last airborne position of a flight on short final to a runway (within 150 m of the centreline, below 700 ft, track within 15 degrees of the runway) after a descent of at least 1 500 ft. It is a landing or presumed landing: a position on the runway afterwards confirmed 47% of US landings in 2024 and 62% in 2025. A go-around is a local altitude minimum below 3 000 ft above the airport, aligned with a runway, preceded by a descent of at least 800 ft and followed by a climb of at least 500 ft with no ground contact. The definition includes missed approaches and climb-outs that begin well above the runway, among them responses to collision-avoidance alerts. It excludes a landing rejected after touchdown. The rate is go-arounds divided by landings plus go-arounds. The detection code was carried over unchanged from an earlier study by the author and was not tuned on these data.

Tight-pair share. For each pair of consecutive landings on the same runway within 400 seconds, the follower's distance from the threshold was taken at the moment the leader was last seen airborne. The measure is the share of pairs in which that distance was under 2.5 NM. A tight pair is not a loss of separation. Spacing of this order is permitted on final at many US airports and is routine when crews maintain visual separation.

Resolution advisory rate. The archive records the resolution advisory (RA) messages broadcast by aircraft collision-avoidance systems (TCAS); such messages have been studied in crowd-sourced data before [13]. An event is the first such message from an aircraft after at least two minutes without one, within 15 NM of a US study airport and below 7 000 ft above it. The rate is events per 1,000 detected approaches, which serves only as a rough scale for traffic: the events include departures and aircraft using nearby smaller airports. Almost no RA messages were received near the European airports, so this analysis has no control group.

Exclusions. An aircraft that made three or more go-arounds at one airport in one day was treated as training, test or flight-inspection activity and all its approaches there that day were removed (216 approaches, including 177 go-arounds), as were 64 approaches by FAA flight-inspection callsigns.

3.4 Design and analysis

The primary estimate is a ratio of rate ratios: the US go-around rate in the shutdown window divided by the US rate on the same weekdays in 2024, divided by the same quantity for the control airports. It was estimated by Poisson regression with a fixed effect for each airport. A value of 1 means US airports changed between the years exactly as the control airports did. Intervals come from resampling whole days 2 000 times, keeping each 2025 day with its 2024 partner, because weather makes go-arounds cluster by day. A second interval resamples airports within the US and control groups. Interval limits vary by about 0.01 between runs.

The decision rules were fixed in advance. A change is detected if the 95% interval excludes 1. No meaningful change is declared if the interval lies wholly inside 0.80 to 1.25. Anything else is inconclusive.

Preregistered secondary analyses were: the same estimate for September (a placebo, since the shutdown had not begun); the shutdown estimate divided by the September estimate

("baseline-adjusted"); separate estimates for 1 October to 6 November (phase A, before the flight reductions), 7 to 16 November (phase B) and 31 October to 9 November (the period of most reported staffing shortfalls); order airports and other US airports separately; each control region alone; airline jets only; and the estimate without the training-flight exclusion. A weather-adjusted model added three airport-day covariates: the share of hours with a ceiling under 1 000 ft or visibility under 3 statute miles, whether any gust reached 25 kt, and whether a thunderstorm was reported.

3.5 Blinding and review

The processing code was built and tested on three days outside the study window. While the 154 study days were processed, only counts of approaches were displayed. The analysis script was committed before use and was run once on the complete data.

Two independent AI reviewers then examined the first draft, one for statistics and reproducibility and one for flight-operations validity. The first rebuilt every airport-day count from the raw files and reproduced every estimate with separate code. Both found problems of interpretation, which this version addresses. The analyses added in response are exploratory and are marked E in the tables and text. The review record is in `AUDIT.md`.

3.6 Departures from the preregistration

1. The weather data were obtained after registration but before any outcome was computed. The covariate definitions and the model were added to the preregistration as a dated addendum at that point (commit `946239f`).
2. The analysis script omitted three registered items: the median follower distance, the airport-resampled interval for the baseline-adjusted spacing estimate, and the correction for multiple testing across airports. All three were computed after the results were known, using the registered definitions.
3. RA events were assigned to the nearest of the 84 registered airports and then dropped if that airport had been excluded, so an event nearer to an excluded airport than to a kept one within 15 NM was lost.
4. All analyses marked E were added after the results were known.
5. Six days failed during processing when the workspace ran out of memory. All six completed on retry and none was dropped.

4 Results

4.1 Go-arounds

In the shutdown window the US rate was 0.12 per 1,000 lower than a year earlier and the control rate 0.22 higher, a difference of −0.34 per 1,000 (Table 1). The ratio of rate ratios is 0.93. The day-resampled interval is 0.76 to 1.13 and the airport-resampled interval is 0.82 to 1.05. Neither excludes 1. Resampling blocks of three consecutive days, to allow for correlation between neighbouring days, widens the interval to 0.74 to 1.18 (E).

The September placebo. The September estimate is 0.78 (0.66 to 0.93). A sound comparison would have given a value near 1. Figure 1 shows why it did not: US rates in September and early October 2024 ran at 5 to 6 per 1,000, well above the same weeks of 2025, while the control airports were level. The gap was broad. Forty of the 50 US airports

Table 1: Go-arounds and approaches by group, year and window. Rates are per 1,000 approaches. The 2024 windows are the same weekdays, 364 days earlier.

Source: `results/results.json`

Window	Group and year	Go-arounds	Approaches	Rate
Shutdown	US 2024	4 062	851 956	4.77
(1 Oct–12 Nov)	US 2025	3 997	860 760	4.64
	Control 2024	1 273	369 735	3.44
	Control 2025	1 423	388 701	3.66
September	US 2024	3 127	579 406	5.40
(1–30 Sep)	US 2025	2 504	587 234	4.26
	Control 2024	908	260 965	3.48
	Control 2025	982	281 238	3.49

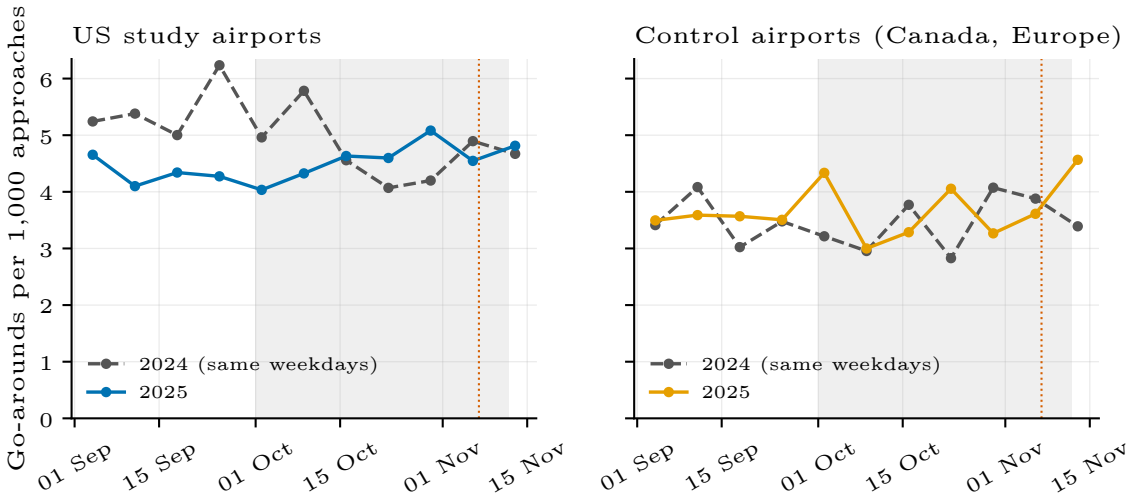


Figure 1: Weekly go-around rate. Solid lines are 2025; dashed lines are the same weekdays in 2024. The shaded band is the shutdown; the dotted line marks the start of the flight reductions on 7 November.

had a lower rate in September 2025 than in September 2024, against 15 of the 26 control airports (E). Leaving out the days around Hurricane Helene (24 to 28 September) gives 0.81 (0.67 to 0.98; E), and the three weather covariates leave the estimate at 0.77 (E).

The reviewers traced part of the gap. Low stratus at the southern California airports was more frequent in September 2024, and an airport such as San Diego is far more sensitive to it than an average weather coefficient allows. Go-arounds linked to collision-avoidance alerts, mostly at San Francisco, numbered 146 in September 2024 and 67 in September 2025 (E). A broad excess in good weather at several large airports, among them Chicago O'Hare, LaGuardia and Dallas/Fort Worth, remains unexplained. A change in practice between the years cannot be excluded, such as less use of visual approaches after the January 2025 collision at Washington.

The gap runs into the shutdown window. Estimated week by week, the ratio is between 0.60 and 0.87 in each of the six weeks from 1 September to 12 October and varies around 1 afterwards. Split in two, the primary window gives 0.67 (0.51 to 0.90) for 1 to 14 October and 1.09 (0.84 to 1.35) for 15 October to 12 November (E). The primary estimate of 0.93 is

Table 2: Preregistered estimates of the ratio of rate ratios for go-arounds, with 95% intervals from resampling days. Values below 1 mean US go-arounds fell relative to the control airports.
Source: `results/results.json`

Analysis	Estimate	95% interval
Primary: shutdown window, all airports	0.93	0.76 to 1.13
September placebo	0.78	0.66 to 0.93
Shutdown, adjusted for September	1.19	0.91 to 1.53
Shutdown, adjusted for weather	0.81	0.68 to 0.95
Phase A, 1 October to 6 November	0.91	0.73 to 1.13
Phase B, 7 to 16 November	0.90	0.65 to 1.21
Peak staffing shortfalls, 31 October to 9 November	1.07	0.92 to 1.26
FAA-order airports only	0.92	0.75 to 1.11
Other US airports only	1.03	0.80 to 1.36
Europe as the only control	0.92	0.75 to 1.13
Canada as the only control	1.00	0.72 to 1.32
Airline jets only	0.93	0.76 to 1.13
Without the training-flight exclusion	0.94	0.77 to 1.14

an average of two weeks that resemble the failed placebo and four weeks slightly above 1.

What the placebo does to the main result. If the September relationship is taken as the baseline, the shutdown estimate is 1.19 (0.91 to 1.53; airport-resampled 0.96 to 1.47). Within 2025 the US rate went from 4.26 per 1,000 in September to 4.64 in the shutdown window, a rise of 9%, while the control rate rose 5%. Within 2024 the US rate fell 12% over the same weeks. The baseline-adjusted figure therefore assumes that 2025 would have repeated the 2024 autumn decline. If early autumn 2024 was the unusual period, the later part of the window is the better guide; if the gap reflects a lasting difference between the years, the baseline-adjusted estimate is. These data cannot say which, and nothing guarantees that the truth lies between the two.

Weather. Low cloud or visibility was the strongest of the three covariates: an airport-day wholly below the threshold had 3.6 times the go-around rate of a day wholly above it. Gusts of 25 kt or more and a reported thunderstorm each multiplied the rate by 1.4. Adjusting for them moves the shutdown estimate from 0.93 to 0.81 (0.68 to 0.95). On the preregistered rule this interval excludes 1. It is not treated as a finding that go-arounds fell. The model applies one set of weather effects to both groups, and the data reject that: low cloud multiplied the rate by 4.6 at US airports and by 2.7 at control airports. With separate weather effects the estimate is 0.87 (0.74 to 1.04; E). The adjustment also acts mainly through the control airports, whose apparent year-on-year rise is an artefact of coverage (below).

Coverage and the control airports. The receiver network saw more in 2025. The share of landings confirmed by a position on the runway rose from 47% to 62% at US airports and from 42% to 55% at control airports (E). Better low-level coverage raises the number of low go-arounds detected. Go-arounds that reached below 200 ft went from 0.66 to 1.09 per 1,000 at the control airports (244 to 422 events) and from 1.02 to 1.11 at US airports. Go-arounds that stayed at or above 200 ft fell in both groups, from 2.78 to 2.58 at control airports and from 3.75 to 3.53 at US airports (E). The whole of the control airports' apparent

Table 3: Exploratory robustness checks added after review. Each row gives the ratio of rate ratios for the shutdown window and, for comparison, the September placebo under the same restriction.
Source: `results/post_review.json`, `results/post_review_ops.json`, `results/supplementary.json`

Analysis (all E)	Shutdown	95% interval	September
Go-arounds reaching 200 ft or higher only	1.03	0.87 to 1.22	0.84
Excluding go-arounds linked to an RA	0.96	0.79 to 1.18	0.79
Both of the above	1.08	0.89 to 1.28	0.86
Excluding 10 airports with a runway in one year only	0.93	0.77 to 1.12	0.80
Excluding 5 airports with large count changes	0.90	0.73 to 1.10	0.77
Weather-adjusted, separate US and control effects	0.87	0.74 to 1.04	–
Adjusted for September and weather	1.07	0.84 to 1.38	–

rise is therefore in the class most dependent on coverage, concentrated at airports such as Zurich, Düsseldorf, Madrid and Gatwick where low-level coverage improved. Restricted to go-arounds at or above 200 ft, the shutdown estimate is 1.03 (0.87 to 1.22). That interval lies inside the 0.80 to 1.25 band. It is an exploratory analysis and its own September placebo is 0.84 (0.69 to 1.01), so it is not offered as a finding of no meaningful change.

Alert-related go-arounds. The detector counts a climb-out in response to a resolution advisory as a go-around. Matching each US go-around to an RA from the same aircraft within three minutes finds 237 such events in the 2024 comparison window and 86 during the shutdown (E), most of them at San Francisco, where a revised visual approach procedure was introduced between the two periods (Section 4.4). Without them the US rate is 4.49 per 1,000 in 2024 and 4.54 in 2025, and the estimate is 0.96 (0.79 to 1.18).

Phases and groups. No phase, airport group or exclusion variant in Table 2 gives an interval that excludes 1 except the weather-adjusted model discussed above. The estimate for the ten days of most reported staffing shortfalls is 1.07 (0.92 to 1.26). In phase B the order airports did not differ detectably from the other US airports (0.85; 0.60 to 1.18). Leaving out any single airport moves the primary estimate only between 0.91 and 0.95. Detected approaches at the order airports were 0.5% higher than in 2024 in phase A and 3.7% lower in phase B (E), so the ordered reductions are visible in the data but small.

Sensitivity achieved. The standard error of the log estimate was 0.102 from resampling days and 0.120 from resampling three-day blocks, against 0.045 if go-arounds were independent events. The realised detectable relative change, on the preregistered definition, is 33% and 40% respectively.

4.2 Variation between airports

Figure 2 shows each airport's year-on-year change. Seventeen of the 76 airports differ between the years at the 5% level on an exact test. That test assumes independent events. Applying it to random halves of a single year's days, where there is no true change, flags 10 to 11 airports on average (95% range 5 to 18; E), so 17 is within what day-to-day clustering alone produces. After the preregistered correction for multiple testing only Washington Dulles remains (22 go-arounds in 15 584 approaches in 2024, 58 in 16 552 in 2025).

The flight-operations review examined the largest changes in the raw data. Most have an explanation unrelated to the shutdown:

- **Boston (rate ratio 1.36).** Approaches detected on runway 27 fell from 7 942 to 436 over the 77 days (E), consistent with a runway closure for safety works reported in

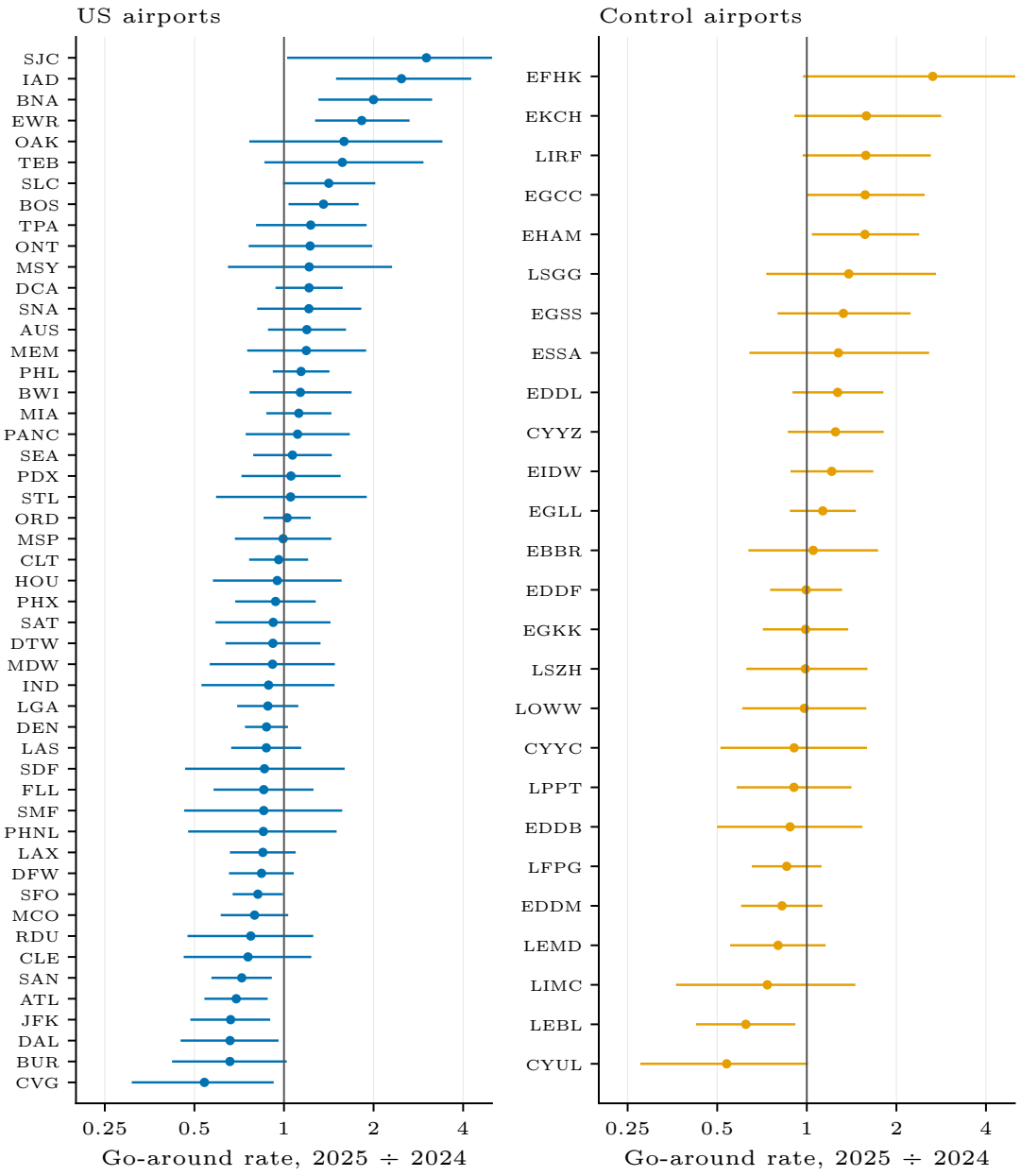


Figure 2: Year-on-year go-around rate ratio in the shutdown window for each airport, with exact 95% intervals that assume independent events and are therefore too narrow. These are single-airport comparisons of 2025 with 2024 and include no control group.

early September 2025 [14]. Arrivals moved to other runways.

- **Nashville (2.00), Cincinnati (0.54), Houston Hobby, Honolulu, Detroit and others.** Ten airports have a runway end that carried at least 4% of landings in one year and almost none in the other (E). Whether each was a closure or a gap in reception, the airport is not comparable across the years.
- **San Diego (0.72).** The rate on runway 27 was similar in the two years. The fall is on runway 09, the low-cloud configuration, which had 1 656 approaches and 87 go-arounds in the 77 days of 2024 and 812 and 27 in 2025 (E).
- **San Francisco (0.82).** Go-arounds linked to a resolution advisory fell from 262 to 73 over the 77 days; other go-arounds rose from 175 to 257 (E).
- **Newark (1.83).** Newark operated under an FAA order limiting hourly arrivals and departures during 2025, with weekend closures of one runway from September [15], which separates it from 2024 for reasons unrelated to the shutdown.

Excluding the ten airports with a runway present in one year only leaves the primary estimate at 0.93 (0.77 to 1.12). The pooled result does not depend on these airports, but the single-airport figures should not be read as effects of the shutdown.

4.3 Spacing on final

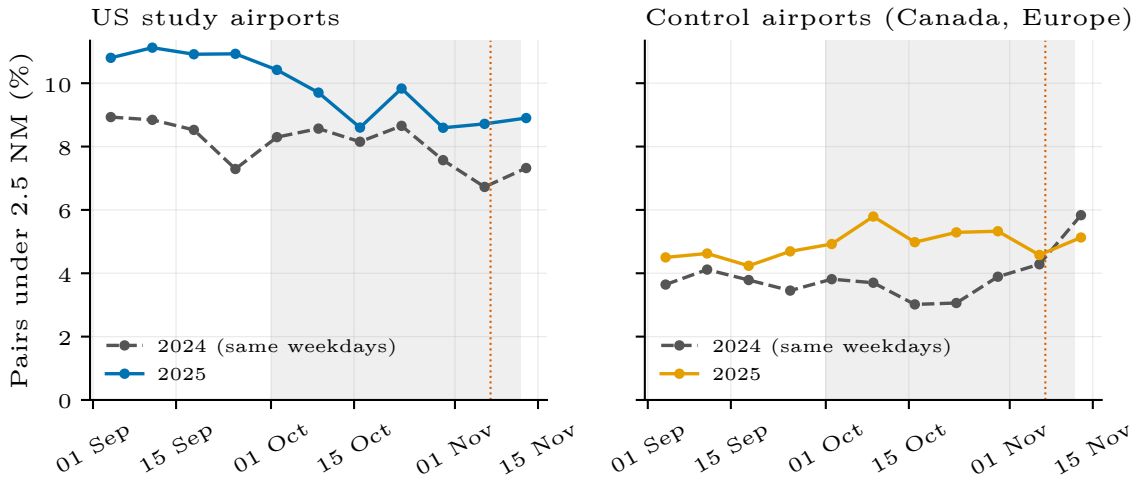


Figure 3: Weekly share of same-runway landing pairs in which the follower was within 2.5 NM of the threshold when the leader was last seen airborne.

The share of tight pairs was higher in 2025 than in 2024 in both groups: 7.9% and 9.2% at US airports, 3.7% and 5.1% at control airports. It rose less at US airports, and the ratio of rate ratios is 0.81 (0.73 to 0.90). The September placebo is 1.03 (0.94 to 1.14). The baseline-adjusted estimate, which the preregistration named as the main spacing result, is 0.78 (0.68 to 0.90 resampling days; 0.65 to 0.91 resampling airports; adjusted $p = 0.001$). The median follower distance at US airports was 4.32 NM in the 2024 window and 4.21 NM in the shutdown window; at control airports 5.00 and 4.75 NM. The median moved towards closer spacing, the opposite direction to the tight-pair result.

On the preregistered analysis this is a detected relative fall in close spacing. It is not interpreted as one, because the measure depends mainly on how low the leading aircraft was tracked. When the leader was last seen below 30 ft, 13.8% of US pairs were tight in both

years. When it was last seen between 30 and 100 ft the share was 3 to 5%, and above 100 ft it was 0.2% or less (E). The share of pairs with the leader tracked below 30 ft rose from 55% to 67% at US airports and from 47% to 59% at control airports. The year-on-year rise in both groups is that shift. Restricted to pairs with the leader tracked below 30 ft, the September estimate is 0.96, the shutdown estimate 0.86 (0.79 to 0.93) and the baseline-adjusted estimate 0.90 (0.78 to 1.03; E). The direction stays negative, the size roughly halves and the interval reaches 1.

Two operational factors also bear on it. Close spacing on final at US airports depends on visual separation, which is used less as autumn weather sets in; the US share declined through autumn 2024 as well (Figure 3). And the operations review found the fall within 2025 concentrated at a few airports, including Boston, which had lost a runway. The spacing data are compatible with slightly wider spacing at US airports during the shutdown and equally with no change.

4.4 Resolution advisories

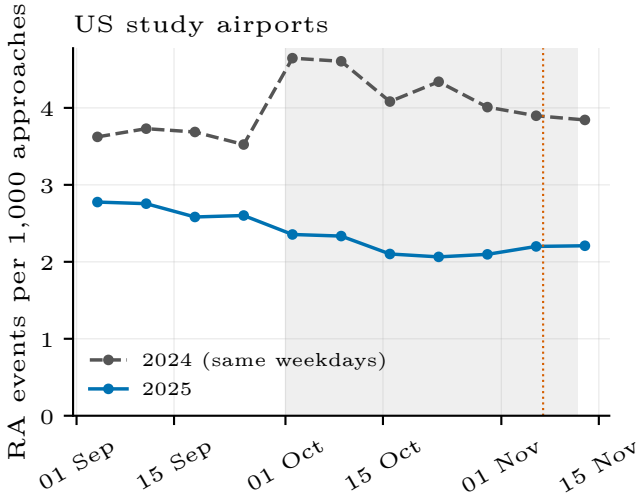


Figure 4: Weekly resolution advisory events per 1,000 detected approaches near US study airports. The step in the 2024 series coincides with a change in the archive's logging of these messages on 26 to 27 September 2024.

RA events near US study airports numbered 2 138 in September 2024 (3.69 per 1,000 approaches) and 3 616 in the 2024 comparison window (4.24); 1 534 in September 2025 (2.61) and 1 883 in the shutdown window (2.19). The preregistered ratio is 0.73 (0.63 to 0.85). Excluding Washington National and Burbank gives 0.76 (0.66 to 0.89), and excluding events whose first message was "Clear of Conflict" gives 0.72 (0.62 to 0.83).

This result is an artefact and is reported only because it was preregistered. Two things produce it.

The archive changed how it logged these messages. The archive held a median of about 6 100 RA messages a day up to 25 September 2024 and about 23 000 a day from 27 September, and the share of messages carrying a position fell from over 99% to about 26% (E). In 2025 the daily volume was steady at about 6 900. The step up in the 2024 series in Figure 4 is that change. It falls at the boundary between the 2024 September baseline and the 2024 comparison window, which is exactly where the preregistered ratio is most sensitive. With the baseline taken from 26 to 30 September in both years, after the change, the ratio is

1.01 (0.78 to 1.34; E).

Two airports dominate, and both changed procedure. San Francisco and Burbank supplied 47% of events in the 2024 comparison window and 17% in the shutdown window (E). At San Francisco the rate fell from 43.7 to 7.4 per 1,000 approaches. The flight-operations review found that these are mostly advisories between aircraft paired on the closely spaced parallel approaches, and a revised visual approach to runway 28R, introduced by one airline in November 2024, is reported to have reduced them [16]. At Burbank the rate fell from 122 to 25 per 1,000; the review found the events clustered over Van Nuys, where the FAA planned to lower the traffic pattern in two steps from summer 2025 to reduce such alerts [17]. Without the two airports the September rates in the two years are almost equal (2.03 and 1.95 per 1,000) and the ratio is 0.83 (0.71 to 0.97; E), which still carries the logging change.

No conclusion about the shutdown is drawn from the RA data. Fewer light aircraft receiving service near large airports could plausibly have reduced advisories during the shutdown, but these data cannot show it.

5 Limitations

- **The comparison years were not equivalent before the shutdown.** The failed September placebo is the main limitation, and the same pattern covers the first two weeks of the shutdown window. The control airports did not remove everything that differed between the years.
- **The control airports are a weak control for US operations.** US airports rely on visual approaches and visual separation far more than European airports do, so the US rate responds more to weather and to runway closures. The control airports remove effects common to the whole dataset, not US operational seasonality. They were not free of disruption either: Nav Canada attributed delays at Toronto on 1 November 2025 to "resource constraints" [18].
- **Sensitivity.** The study could detect a relative change of about a third. A rise of 10 or 20% in go-arounds would not have been detected reliably.
- **Receiver coverage improved between the years**, and the direction of its effect is known: more low go-arounds detected, more landings confirmed, leaders tracked lower. About 80% more positions per aircraft were kept near the study airports in 2025. The restriction to go-arounds at or above 200 ft addresses this only in part.
- **Go-around detection was not validated against a reference for these airports.** The operations review found that about 90% of detected US go-arounds were followed by a landing by the same aircraft at the same airport within 30 minutes, which supports the detector's specificity. Published work finds that detection from crowd-sourced data tends to miss go-arounds [19].
- **The reason for each go-around is unknown**, as is who initiated it.
- **Runway availability was not controlled.** Ten airports had a runway in one year only, and closures were not checked against official records.
- **The two years differ in other ways.** Hurricane Helene fell in the September 2024 baseline and Hurricane Milton in the 2024 comparison window. The UPS accident at Louisville occurred on 4 November 2025. From 7 November the flight reductions, the general aviation restrictions and the withdrawal of some services to VFR traffic changed the traffic itself.

- **Staffing was not measured.** The study compares periods, not facilities with and without shortfalls.
- **The literature search was not exhaustive**, and several operational explanations supplied by the review rest on trade or news reports that were not checked against primary records.
- **The preregistration is time-stamped only by a public repository commit.**

6 What the study does and does not show

It shows that, across 50 US airports and about 860 000 approaches during the shutdown, the rate at which jets discontinued approaches was close to the rate a year earlier, and that no version of the analysis found a clear increase. It shows that this holds when low go-arounds, alert-related go-arounds and airports with runway or coverage changes are set aside. It also shows how easily open flight-track data produce convincing false signals: two preregistered secondary analyses returned statistically clear falls that were artefacts of the data source and of local procedure changes.

It does not show that the shutdown had no effect on safety. The formal result is inconclusive. The check designed to confirm that the comparison was fair failed, and a moderate rise in go-arounds cannot be excluded once that failure is allowed for; neither can a fall. A steady go-around rate is not a safety verdict in either direction. The study did not measure the events the FAA cited, which were voluntary reports and unspecified safety data, and it did not measure losses of separation, runway incursions, communication errors or controller fatigue. It says nothing about any single airport.

What would settle it. Four things would sharpen the answer: the FAA's own counts of the events it referred to; a facility-level comparison that uses the dates and places of staffing shortfalls; a third comparison year, which would show whether autumn 2024 or autumn 2025 was the unusual one; and reprocessing with the 2025 data thinned to the 2024 density, so that coverage no longer differs between the years.

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Competing interests. The author is the founder of CitadelAero, which builds AI tools for aviation.

Disclaimer. The views expressed are the author's own and do not represent those of any employer or regulator. Nothing here is operational, legal or regulatory advice.

Materials. The preregistration, code, airport-day data, results, verification log and audit record for this paper are at `https://github.com/JamesLeCuirot/Research/tree/main/papers/08-shutdown-safety-indicators`.

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